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ODD END SEMESTER EXAMINATION DECEMBER - 2024

Name of the Course: B. Tech (Electronics & Communication Engg. with specialization in IoT) and B. Tech (Electronics & Communication Engg. with specialization in ESR)

Semester: IIIrd

Name of the Paper: Fundamentals of Computer Organization

Paper Code: **TEC-359**

Time: 3 Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question.
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks.

Q1	(20marks)	CO1
(a)	Difference between Computer architecture and Computer Organization.	
(b)	Draw and Explain the functional components of Computer.	
(c)	Differentiate between RISC and CISC.	
Q2	(20 marks)	CO2
(a)	Explain the different addressing modes in terms of algorithm, advantage, disadvantage and example.	
(b)	Write a program that can evaluate following expression: $X = (A * B) + (C * D)$ using three address, two address, one address and zero address instructions.	
(c)	Explain computer instruction and datatypes used.	
Q3	(20 marks)	CO3
(a)	Explain the IEEE 754 format for 64 bits and arrange the number $(75.125)_{10}$ in the mentioned IEEE 754 format.	
(b)	Explain Booth's multiplication algorithm and hence multiply $(-5)*(-7)$	
(c)	Explain the binary division algorithm with example.	
Q4	(20 marks)	CO4
(a)	Differentiate between Hardwired control unit and Microprogrammed control unit	
(b)	Explain registers used with example.	
(c)	Explain Control unit of the processor.	
Q5	(20 marks)	CO5 and CO6
(a)	Explain Memory Hierarchy Design and its Characteristics	
(b)	Explain the concept of Cache memory briefly.	
(c)	Explain the difference between memory mapped I/O and I/O mapped I/O.	

